

McGregor Fellowship 2026

Teaching Ancient Wisdom with Modern AI: Evaluating LLMs in Early Christian Studies

Student Working Plan | May 11 – July 21, 2026

Faculty Mentor: Prof. Eric Araújo
Department of Computer Science | Calvin University
Ian Hawthorne & Nathan Holtrop

10 weeks Program Duration	9 wks work After 1-wk Break	40 hrs/wk Max per Student	Jul 17 Final Presentation
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This document is to be shared with McGregor Fellows at the first meeting on or before May 11. It sets out the reading plan, weekly schedule, task division, and deliverables for the full program period. Fellows should read it carefully before the first day.

Last update: May 2, 2026

1. Program Overview

The McGregor Fellowship pairs undergraduate researchers with faculty mentors for a 10-week summer research experience. This project investigates how accurately large language models (LLMs) represent women in early Christian history — a domain where AI systems are known to hallucinate, minimize, and erase female figures.

Two students will work full-time (up to 40 hours per week) from May 11 through July 21, 2026, with a scheduled one-week break from June 15–19. The program culminates in a public research presentation on July 17.

1.1 Program Calendar

Week	Dates	Status	Focus
1	May 11–15	Active	Orientation, workspace setup, reading begins
2	May 18–22	Active	Historical foundation: broad church history reading
3	May 25–29	Active	Primary sources + corpus finalization
4	Jun 1–5	Active	Scholarly works + prompt design begins
5	Jun 8–12	Active	Prompt refinement + API pipeline setup
BREAK	Jun 15–19	Break — no work	McGregor program break (no work)
6	Jun 22–26	Active	Data collection: GPT-4 / Claude
7	Jun 29 – Jul 3	Active	Data collection: remaining models (Jul 4 holiday)
8	Jul 6–10	Active	Validation + error analysis
9	Jul 13–17	Active	Final synthesis, report, presentation (Jul 17)
Wrap	Jul 18–21	Partial	Documentation, archiving, wrap-up (24h)

July 4 (Independence Day) falls on Saturday in 2026 — no impact on the work week. Week 7 is a full 40-hour week.

1.2 Project Focus: Two Primary Figures

While the broader question bank covers 15 female figures across the 1st–5th centuries, the reading plan and data collection this summer centers on two fourth-century women who are exceptionally well-documented:

- **St. Macrina the Younger (c. 327–379)** — philosopher, theologian, monastic founder; elder sister of Basil of Caesarea and Gregory of Nyssa (Cappadocian circle).
- **Olympias the Deaconess (c. 361–408)** — deaconess, philanthropist, monastic leader; spiritual correspondent of John Chrysostom in Constantinople.

These figures provide contrasting geographical contexts (Cappadocia vs. Constantinople), contrasting documentation types (hagiography vs. epistolary), and contrasting institutional roles (monastic founder vs. ecclesiastical deaconess), making them ideal anchors for a first evaluation cycle.

2. Reading Plan: Books, Page Counts & Reading Time

The following tables present all assigned and recommended books with estimated page counts, difficulty levels, reading speeds, and total reading hours. These estimates assume undergraduate-level academic reading pace. Students should log their actual reading hours each week.

Page counts are estimates based on publisher series norms. Verify against your physical copies when they arrive. Reading hours account for active reading with note-taking, not skimming.

Section A — Required Reading (one copy per student)					
Title & Author	Pages*	Level	pp/hr	Hours	Weeks
The Story of Christianity, Vol. 1 Justo L. González (HarperOne, 2010)	~400	Accessible	35	~11.5h	1–2
Desert Fathers and Mothers Christine Valters Paintner (SkyLight Paths, 2012)	~176	Accessible	30	~6h	1–2
The Forgotten Desert Mothers Laura Swan (Paulist Press, 2001)	~208	Accessible	28	~7.5h	2–3
The Life of Saint Macrina Gregory of Nyssa, trans. Kevin Corrigan (Wipf & Stock, 2005)	~100	Moderate (primary source)	15	~6.5h	3
Letters to Saint Olympia John Chrysostom, trans. David C. Ford (SVS Press, 2017)	~130	Moderate (primary source)	15	~8.5h	3–4
Christian Women in the Patristic World Lynn H. Cohick & Amy Brown Hughes (Baker Academic, 2017)	~352	Academic	22	~16h	4–5
SECTION A TOTAL	~1,366	—	—	~56h	Wks 1–5

Section B — Recommended Reading (1 copy shared; divide by student)					
Title & Author	Pages*	Level	pp/hr	Hours	Weeks
Women in the Early Church Elizabeth A. Clark (Liturgical Press, 1983)	~272	Academic	22	~12.5h	4–5
Macrina the Younger, Philosopher of God Anna M. Silvas (Brepols, 2008)	~280	Dense academic	18	~15.5h	5–7
Virgins of God: The Making of Asceticism Susanna Elm (Oxford, 1996)	~450	Dense academic	18	~25h	5–8 (selective)
SECTION B TOTAL (all books, if read fully)	~1,002	—	—	~53h	—
SECTION B TOTAL (divided between students)	≈501 each	—	—	~27h each	Wks 4–8

2.1 Total Reading Commitment per Student

Category	Pages	Hours	When
Section A — Required (both students read all)	~1,366	~56h	Weeks 1–5
Section B — Each student reads half, shares notes	~500	~27h	Weeks 4–8
TOTAL per student (estimated)	~1,866	~83h	10 weeks

~83 hours of reading out of 360 total working hours = approximately 23% of total time. Reading is heaviest in weeks 1–3 and serves as the knowledge foundation for all validation work in weeks 7–8.

2.2 Section B Division Between Students

Since Section B books are shared (one physical copy between two students), each student takes primary responsibility for reading and summarizing different books, then sharing detailed notes:

- **Ian Hawthorne:** Women in the Early Church (Clark, ~12.5h) + Macrina the Younger, Philosopher of God (Silvas, ~15.5h)
- **Nathan Holtrop:** Virgins of God (Elm) — read chapters 1–5 and chapters relevant to Macrina and Olympias (~25h for selective reading)
- **Both students:** Share written summaries and key quotations for all Section B books by the end of Week 7

Virgins of God (Elm) is a 450-page dense academic monograph. Students are NOT expected to read it cover to cover; instead, Nathan reads the introduction, chapters directly relevant to Macrina and Olympias, and the conclusion. Prof. Araújo will identify the priority chapters in Week 1.

3. Reading Difficulty Guide

Understanding what ‘difficulty level’ means helps students pace their reading and know what to expect. The three levels used in this plan are:

Level	Vocabulary	Sentence Structure	Strategy
Accessible	Common English; minimal jargon	Clear, narrative flow; short chapters	Read continuously; 30–40 pp/hr
Moderate (incl. primary sources)	Specialist vocabulary; translated ancient prose	Longer sentences; unfamiliar cultural context	Slow down, take notes, use glossary; 15–25 pp/hr
Dense Academic	Heavy use of German/French secondary lit. refs; archaic Greek/Latin terms	Long chapters; dense footnotes; assumes prior expertise	Read actively; take structured notes; re-read key passages; 15–20 pp/hr

Primary sources (Chrysostom’s letters, Gregory of Nyssa’s *Life of Macrina*) require especially careful reading because they are translations of 4th-century Greek texts. Read slowly. Every claim in these texts can become an evaluation question or a ground-truth answer.

4. Week-by-Week Working Plan

Each week is 40 hours. The plan is structured around five phases that mirror the project proposal. McGregor community meetings (required for all fellows) occur each week and are already built into the schedule.

McGregor community meetings run approximately 2 hours per week. This time is part of the 40-hour week and should be logged on the timesheet.

Week 1 | May 11–15 | Orientation & Setup

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Begin <i>The Story of Christianity Vol.1</i> (≈70 pages)	Meet with Prof. Araújo: project goals, methodology, literature review Set up: GitHub repo, shared drive, documentation system Review existing <code>literature_review.md</code>	GitHub repo initialized Shared documentation folder created
Wed–Thu	16h	Continue González (≈70 pages more)	Set up API access: OpenAI, Anthropic, Grok accounts Install Python environment; test batch-collection notebooks Review the 42-question bank	API keys working Notebooks running
Fri	8h	Begin <i>Desert Fathers & Mothers</i> (Paintner)	McGregor community meeting (2h) Reading log + reflection notes submitted	Week 1 reading log submitted

Week 2 | May 18–22 | Historical Foundation

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Finish González Ch. 1–15 (complete if not done) Finish Paintner	Literature review: identify key scholarly debates on Macrina and Olympias Begin preliminary summary profile: Macrina the Younger	Macrina profile draft (1–2 pages)
Wed–Thu	16h	Begin <i>The Forgotten Desert Mothers</i> (Swan) (≥80 pages)	Begin preliminary summary profile: Olympias Identify authoritative primary and secondary sources for validation	Olympias profile draft (1–2 pages)
Fri	8h	Continue Swan	McGregor community meeting (2h) Sync meeting with Prof. Araújo: review profiles, answer questions	Both figure profiles submitted

Week 3 | May 25–29 | Primary Sources + Corpus Finalization

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Finish Swan Begin Life of Saint Macrina (Corrigan)	Finalize list of figures and events for evaluation (with Prof. Araújo) Confirm scope: Macrina + Olympias as primary; broader 42-question bank as secondary	Finalized corpus document
Wed–Thu	16h	Finish Life of Macrina Begin Letters to Olympia (Chrysostom)	Create detailed fact-check documents for Macrina and Olympias Key dates, roles, sources, common misattributions, scholarly debates	Fact-check sheet: Macrina Fact-check sheet: Olympias
Fri	8h	Continue Letters to Olympia	McGregor community meeting (2h) Begin researching prompt engineering best practices	Week 3 reading log submitted

Week 4 | June 1–5 | Scholarly Works + Prompt Design

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Finish Letters to Olympia Begin Christian Women in the Patristic World (Cohick & Hughes) Ian: Begin Women in the Early Church (Clark)	Draft 15–20 benchmarking prompts for Macrina (biographical, chronological, theological, contextual) Test prompts on 1–2 LLMs informally	First draft: Macrina prompt set
Wed–Thu	16h	Continue Cohick & Hughes Nathan: Begin Virgins of God, Elm (priority chapters)	Draft 15–20 benchmarking prompts for Olympias Draft adversarial prompts (false-premise questions) Compile annotation rubric first draft	First draft: Olympias prompt set Rubric v0.1
Fri	8h	Continue reading	McGregor community meeting (2h) Meeting with Prof. Araújo: prompt review + rubric feedback	Revised prompts after feedback

Week 5 | June 8–12 | Prompt Finalization + API Pipeline Setup

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Finish Cohick & Hughes Continue Section B (as assigned)	Run pilot prompts on GPT-4 and Claude; evaluate quality Refine prompts: fix ambiguities, strengthen adversarial questions	Final prompt set (all models)

Day	Hours	Reading	Technical / Research	Deliverable
			Finalize prompt set with Prof. Araújo	
Wed–Thu	16h	Continue Section B reading	Set up batch-collection pipeline (Python/API notebooks) Test pipeline with 5–10 prompts; verify output structure Finalize annotation rubric (v1.0) with all dimensions	Pipeline running end-to-end Rubric v1.0 finalized
Fri	8h	Continue Section B reading	McGregor community meeting (2h) Final pre-break meeting: review everything before break	Pre-break status report submitted

BREAK: June 15–19. No work required. Enjoy the rest. Return refreshed on June 22.

Week 6 | June 22–26 | Data Collection Phase 1

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Continue / finish Section B notes	Query GPT-4o with complete prompt set (all figures) Document all responses: model, prompt ID, figure, timestamp Organize into database/spreadsheet	GPT-4o dataset complete
Wed–Thu	16h	Section B notes continued	Query Claude (claude-opus-4 or equivalent) with full prompt set Document responses; flag any technical errors or rate limits Begin cross-model comparison: preliminary observations	Claude dataset complete
Fri	8h	Section B notes	McGregor community meeting (2h) Data quality review with Prof. Araújo	2-model dataset reviewed and clean

Week 7 | June 29 – July 3 | Data Collection Phase 2 + Validation Begins

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Reference Elm/Silvas as needed for validation	Query Grok and/or one open-source model (Llama 3, Mistral) with full prompt set Complete data collection from all planned LLMs Finalize and index all responses	All LLM datasets complete and indexed

Day	Hours	Reading	Technical / Research	Deliverable
Wed–Thu	16h	Reference primary sources as needed	Begin systematic validation: compare LLM outputs vs. authoritative sources Apply error taxonomy: factual inaccuracy, anachronism, conflation, hallucination, omission, gender erasure Validate first 20% of responses	Validation: 20% complete
Fri	8h	Reference primary sources	McGregor community meeting (2h) Share Section B notes with each other (swap summaries)	Section B notes exchanged

Week 8 | July 6–10 | Comprehensive Validation + Error Analysis

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h	Reference books as needed	Complete validation of all LLM responses (60–100% remaining) Categorize all identified errors using error taxonomy Compute accuracy rates per model, per figure, per question difficulty	Validation complete Error taxonomy spreadsheet
Wed–Thu	16h		Identify patterns: which question types hallucinate most? Which models? Look for gender-bias patterns: erasure, minimization, anachronistic framing Prepare preliminary findings document	Preliminary findings report (2–3 pages)
Fri	8h		McGregor community meeting (2h) Meeting with Prof. Araújo: review findings, prepare for final reporting	Findings review complete

Week 9 | July 13–17 | Final Synthesis + Public Presentation

Day	Hours	Reading	Technical / Research	Deliverable
Mon–Tue	16h		Aggregate findings across all models, figures, and question types Statistical analysis: accuracy percentages, error distributions, model comparisons Create visualizations: charts, tables, heatmaps of LLM performance	Final analysis and visualizations
Wed–Thu	16h		Write final research report Prepare presentation slides for July 17 McGregor showcase Rehearse presentation (target: 15–20 min + Q&A)	Final report draft Presentation slides

Day	Hours	Reading	Technical / Research	Deliverable
Fri Jul 17	8h		McGregor community meeting (2h) 🎓 Public Research Presentation	Presentation delivered McGregor program complete

Wrap-up Week | July 18–21 | Documentation & Archiving (~24h)

The final four days are for archiving the project, finalizing documentation, and handing off materials for future phases. Prof. Araújo will host a wrap-up meeting to reflect on the summer and discuss publication plans.

- Archive all data, code, and documentation in the shared repository
- Write brief individual reflection: what you learned, what surprised you
- Prepare any publication-ready materials if applicable
- Complete McGregor Fellow evaluation (required by the program)

5. Task Division Between Students

Both students participate in all phases of the project. However, to maximize efficiency, certain responsibilities are assigned as ‘lead’ roles. The lead student takes primary ownership of a task but always keeps the other student informed.

Task	Ian Hawthorne (Lead)	Nathan Holtrop (Lead)
API setup & pipeline configuration	✓ Lead	Support
Data collection & organization	Lead: Macrina figures	Lead: Olympias figures
Prompt design	Both contribute equally	Both contribute equally
Section A reading (all required books)	Both read all books	Both read all books
Section B — Clark + Silvas	✓ Lead (reads + summarizes)	Receives summary
Section B — Elm (priority chapters)	Receives summary	✓ Lead (reads + summarizes)
Fact-check sheets: Macrina & related	✓ Lead	Support + review
Fact-check sheets: Olympias & related	Support + review	✓ Lead
Validation & error coding	Both contribute equally	Both contribute equally
Statistical analysis & visualizations	✓ Lead	Support
Final report writing	Both contribute equally	Both contribute equally
Presentation preparation	Both contribute equally	Both contribute equally

The lead/support split is flexible and should be renegotiated with Prof. Araújo if students have specific backgrounds or preferences (e.g., one student is a CS major, the other a theology or history major). The goal is to play to strengths while ensuring both students develop competency in all dimensions.

6. Key Deliverables & Milestones

Week	Due Date	Deliverable
1	May 15	GitHub repo initialized; shared workspace set up; reading log started
2	May 22	Macrina and Olympias figure profiles (1–2 pages each); Week 2 reading log
3	May 29	Finalized evaluation corpus; fact-check sheets for Macrina and Olympias
4	Jun 5	First draft prompt set (~40 prompts); annotation rubric v0.1
5	Jun 12	Final prompt set (all models); annotation rubric v1.0; batch pipeline tested
6	Jun 26	GPT-4 and Claude datasets complete and documented
7	Jul 3	All LLM datasets complete; validation 20% done; Section B notes exchanged
8	Jul 10	Full validation complete; error taxonomy spreadsheet; preliminary findings report
9	Jul 17	FINAL: Research report + presentation delivered at McGregor showcase

7. McGregor Program Requirements

The following are non-negotiable requirements of the McGregor program that apply to all fellows and mentors:

- **Weekly community meetings:** All fellows attend the required weekly McGregor meeting (approximately 2 hours). This is part of your 40-hour week.
- **Weekly timesheets:** Students submit timesheets weekly. Prof. Araújo approves them. Log your hours accurately, including reading, research, meetings, and technical work.
- **Maximum hours:** Students may NOT work more than 40 hours per week. This is a strict program rule.
- **Presence:** Students are expected to be in Grand Rapids for the full program period (May 11–July 21) and at all McGregor meetings, even if work arrangements are adjusted.
- **Break week:** June 15–19 is an official break. No work is expected or logged during this week.
- **Final presentation:** All fellows present their research on July 17. Attendance is mandatory. Prof. Araújo is required to attend the full session.
- **End-of-term evaluation:** Prof. Araújo will evaluate each student's performance at the end of the program. Students will also complete a self-evaluation.

McGregor mentor presentations: Faculty mentors are asked to give a 30-minute presentation on their scholarly journey during the weekly community meetings. Prof. Araújo will do this during one of the meeting weeks. Students should attend and engage actively.

7.1 Working Norms

To make this a positive 'working with' experience (rather than 'working for'), the following norms apply:

- **Weekly one-on-one with Prof. Araújo:** Every Friday, in addition to the McGregor community meeting. Agenda: progress check, questions, reading discussion, planning for next week.
- **Reading logs:** Both students maintain a shared reading log noting pages read, key insights, and questions. This is submitted weekly.
- **Teams/email response time:** Respond to messages from Prof. Araújo within one business day.
- **Intellectual engagement:** You are expected to bring your own questions, observations, and ideas. If you notice something interesting in an LLM response or a primary source, flag it — these become research insights.

8. Estimated Hour Budget per Student

The following is a rough breakdown of how 360 total hours (9 weeks × 40h) are spent. These are approximations; actual allocation will vary by week and by student.

Activity	Weeks	Hours	% of Total
Reading (Section A + half of B)	1–5	~83h	~23%
Orientation, meetings, documentation	1, 9	~30h	~8%
Figure profiles, fact-check sheets, corpus work	2–3	~25h	~7%
Prompt design and refinement	3–5	~30h	~8%
API pipeline setup and testing	4–5	~20h	~6%
Data collection (all models)	6–7	~50h	~14%
Validation and error coding	7–8	~60h	~17%
Statistical analysis and visualization	8–9	~25h	~7%
Final report and presentation	9	~25h	~7%
Wrap-up / archiving	10	~12h	~3%
TOTAL		~360h	100%

A note to students: *This project sits at an unusual intersection — computer science, early Christian history, and questions about how AI systems perpetuate or correct historical bias. The women you will read about — Macrina, Olympias, and the others in the question bank — were real people with real intellectual and spiritual legacies that have been minimized for centuries. Your work this summer contributes to a better understanding of where AI systems succeed and where they fail when it comes to figures like these. That is meaningful research, and I (Prof. Araújo) am glad to have you as partners in it.*

Prof. Eric Araújo | Calvin University | Computer Science
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